S Nikhil Ram
Github : niksram
Mobile : (+91) 9972124102
Email : nikhilsram.off@gmail.com
Web : niksram.github.io

Education

- **PES University** — Bengaluru
  Bachelor of Technology in Computer Science and Engineering; ***Curr. CGPA: 9.51/10.00*** — *Aug. 2018 – Present*
- **Vidya Mandir Ind. PU College** — Bengaluru
  Pre-University Education; ***Per.: 93.7%*** — *July 2016 – May 2018*
- **Sindhi High School** — Bengaluru
  Class 1-10 in CBSE; ***CGPA: 10.0/10.0*** — *June 2006 – March 2016*

Skills

- **Programming Languages**: Python, C, C++, Java, Shell-Scripting, JavaScript, SQL, HTML-CSS
- **Technologies**: BPFTrace, Ftrace, DeathStarBench, Docker, SpringMVC, AngularJS, Django, Hadoop, PySpark, AWS-Hosting, Git, Java-Collections, STL(C++)

Research

- **MiSeRTrace - Kernel-level Request Tracing for Microservice Visibility** — PES University, Bengaluru
  Undergraduate Final Year Thesis — *Jan 2021 - Present*
  - MiSeRTrace (MicroService Request Trace) is a tool developed to trace the end-to-end path of requests entering a microservice application at the kernel space without instrumentation or modification of the application code.
  - In addition to its primary functionality, it supports tracing user-enabled kernel trace-points and events provided by BPFTrace and Ftrace and isolating them for each request.
  - Observability at the comprehensiveness of kernel space allows breakdown of steps involved in activities such as Network Transfers, IO Operations and handling Interrupts at the granularity of kernel threads.
  - *Technologies - Linux Kernel, BPFTrace (eBPF), FTrace, DeathStarBench, Docker.*

Experience

- **Advanced Micro Devices, Inc. (AMD)** — Bengaluru
  Server Performance Engineer — *Dec 2021 - Present*
  - Analysed SPEC2017rate workloads and performance uplifts in relation with architectural dependencies - CPU frequency, memory frequency, uncore frequency, LLC cache scaling, SMT bindings, pre-fetchers, etc and unarchitectural dependencies - compiler version, static vs dynamic binaries, types of Malloc used, compiler flags, etc.
  - Benchmarked database workloads - Mongo, Cassandra, Memcached, Redis, MySQL, Aerospike, etc to analyse performance uplifts related to CPU bindings, server-client instance scaling, etc with the use of internal tools to trace and debug the cause.
  - *Technologies - SPEC2017, Database Workloads, Internal Tools, Perf.*

- **Morgan Stanley** Bengaluru
  Summer Technology Analyst *May 2021 - July 2021*
  - Developed a Dynamic Real-Time Analytical tool that provides reporting and analytical capabilities to its end users based on a simple search.
  - The search is performed by constructing dynamic database queries, service calls to various endpoints, parsing metadata files and the consolidated data is represented in an interactive dashboard.
  - *Technologies - SpringMVC (JAVA), AngularJS, UI Bootstrap, Tomcat Server Deployment.*
- **Center for Cloud Computing and Big Data** PES University, Bengaluru
  Research Intern *June 2020 - July 2020*
  - Worked with a team to develop a Cloud Based Single Page Web Application for the Department of Computer Science, PES University for evaluating final year projects.
  - The Web-Application was Dockerized into containers and deployed on an Amazon EC2 Instance.
  - *Technologies - DjangoREST, JWT, HTML-CSS, JavaScript, Docker.*
  - **Presented at IEEE-CCEM** *(IEEE International Conference on Cloud Computing in Emerging Markets).*

Projects

- **YACS (Yet Another Centralised Scheduler)**: Simulated a scheduling framework to manage and allocate the resources of a cluster to the incoming jobs. *Python, Parallel-Processing*
- **Greenest Parts of Bengaluru**: Hadoop Map-Combine-Reduce was used to determine the greenest parts of Bengaluru. The process was run on a Pseudo-Distributed Apache-Hadoop cluster. *Python, Hadoop-MRJob, OpenCV*
- **Simulation of Multi Parametric Natural Selection**: Literature survey on Natural Selection and developed a model using conventional Linear Algebra and Monte-Carlo Techniques. *Literature Survey, Python*
- **Mini Projects**:
  - **Cache Mapper Simulator** to simulate cache handling techniques at hardware level.
  - **IntAL C Library** to perform arithmetic operations on whole numbers of the order 10^{1000}.
  - **Best Fit Memory Management** in a given Page of memory.
  - **RegEx Engine** in C to match strings.
  - **SQL Version Control Management System Schema** to simulate functioning of Github.

  C, C++, STL, SQL

Achievements

- **Prof. C N R Rao Merit Scholarship**: Semester 1 and 2, *Offered to the top 20% students of PES University.*
- **Prof. M R Doreswamy Merit Scholarship**: Semester 3, 4, 5, 6 *Offered to the top 20% students of PES University.*
- **Certificate of Merit from Central Board of Secondary Education**: Securing A1 in all the subjects of class 10.

Participation

- **e-Yantra - 2018**: Worked with a team in the e-Yantra Robotics Competition, a project initiative by IIT Bombay and qualified for the pre-finals.
- **Google Hash Code**: Participated in the years 2019, 2020 and 2021.

Extra Curriculum

- **Trinity College of London**: Graded Examination in Piano **Grade 1-5**.
- **Trinity College of London**: Graded Examination in Music Theory **Grade 1-6**.
- **Carnatic Music (Vocal) Junior Grade**: Qualified First Class.